

An Overview of Fuel

Fuel is an open source deployment and management tool for OpenStack, and is a community-driven initiative. It has a GUI for the deployment and management of OpenStack, using its many components like Nailgun, Astute, Keystone, Postgres, etc.



Fuel is an open source tool for deploying OpenStack with high availability in multiple racks of servers or for a proof-of-concept environment with one or two servers.

Fuel has many key features, some of which are listed below:

- Management of multiple OpenStack environments
- Auto discovery of new nodes and their hardware
- IP address management
- HA and non-HA deployment
- Centralised logging
- Support for popular Software Defined Networks (SAN), storage and other features through external plugins
- Support for multiple distros
- Industry standard high-availability configuration
- Support for multiple hypervisors

The components of Fuel

Nailgun: This is a RESTful application that can create environments, change settings and configure networks; it uses Postgres to store data and uses AMQP to interact with workers.

Astute: This process executes instructions given by Nailgun, reads the information from AMQP put up by Nailgun, and applies it accordingly on the managed

nodes. This interacts with Cobbler, Puppet and other components for deployment.

Keystone: This is for authenticating and authorising users.

Postgres: This is the database used for storing all the environment and configuration details.

Cobbler: This is a popular open source OS deployment tool, which manages TFTP and DHCP services.

Astute interacts with Cobbler to create a profile for each managed node.

Puppet: Most of the deployment is done through Puppet manifests, which are synced to all managed nodes and executed by the MCollective agent running on each node.

rsync: This runs as a daemon, which helps sync all deployment scripts with managed nodes. Scripts are pulled before the deployment by managed nodes.

Configuring and installing Fuel

In this set-up, I will use three nodes for running OpenStack services and two nodes for compute.

Prerequisites: Fuel uses five different networks for five different purposes, so the data centre network has to be configured accordingly.

The Fuel master node (the one on which Fuel will be